# Chemistry of Nucleotides

# Learning Objectives

1. Nitrogenous bases
2. Nucleosides
3. Nucleotides

Nucleotides are low molecular weight, nitrogenous compounds

They are essential for life

They perform some vital functions in all living organisms

**Nucleotides act as:**

- Monomers of nucleic acids
- High-energy compounds
- Second messengers
- Coenzymes
- Metabolic intermediates

# Nucleotides as monomers

Nucleotides are the monomeric units of nucleic acids

Ribonucleotides contain the sugar, ribose

Deoxyribonucleotides contain the sugar, deoxyribose

# Nucleotides as high-energy compounds

Some nucleotides are high-energy compounds

An example is adenosine triphosphate (ATP)

ATP is called the energy currency of cells

It is formed in exergonic reactions, and is

# Nucleotides as second messengers

Some nucleotides control biochemical and physiological processes by acting as second messengers for hormones

Examples are cyclic adenosine mono- phosphate (cAMP) and cyclic guanosine monophosphate (cGMP)

# Nucleotides as coenzymes

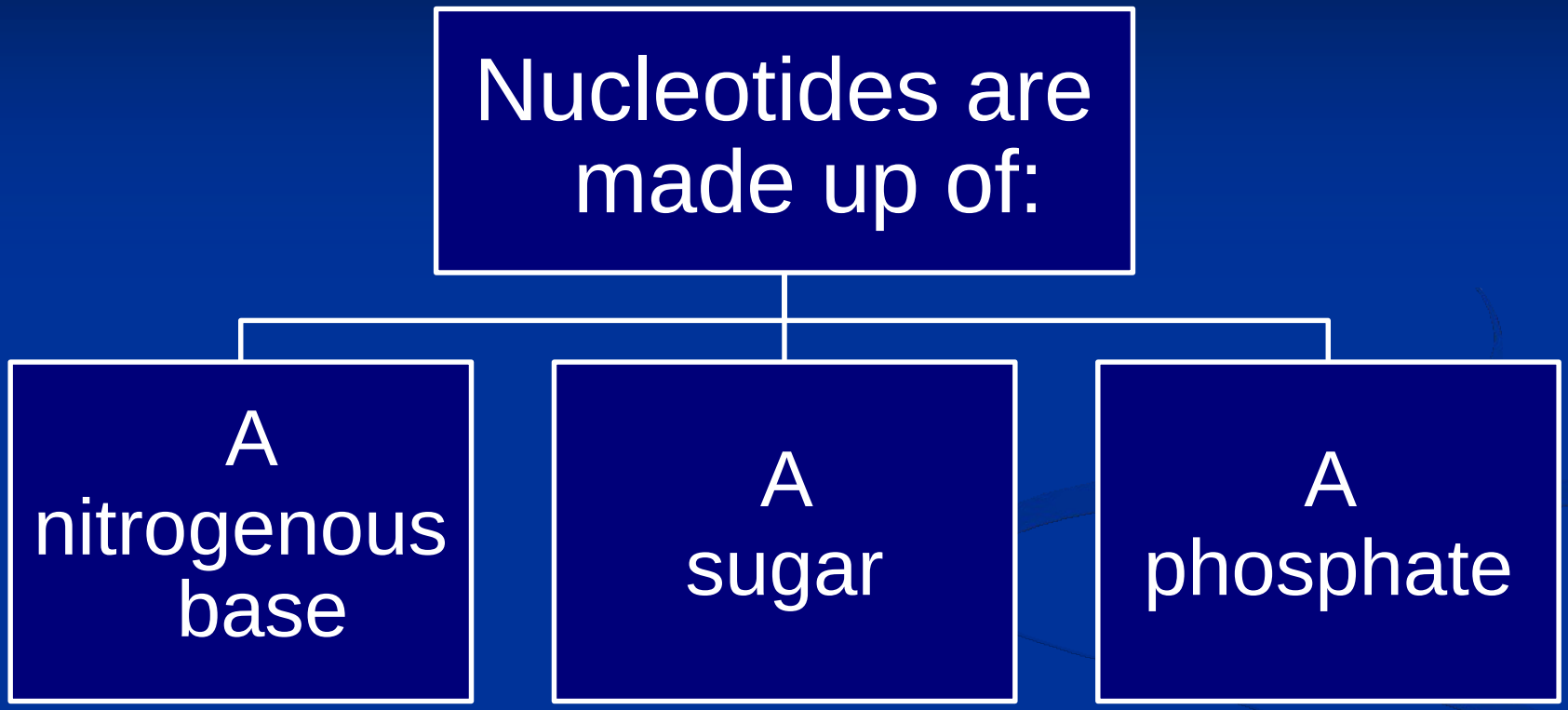
- Many nucleotides act as coenzymes:
  - Flavin adenine dinucleotide (FAD)
  - Nicotinamide adenine dinucleotide (NAD)
  - Nicotinamide adenine dinucleotide phosphate (NADP)

# Nucleotides as metabolic intermediates

Some nucleotides act as intermediates in metabolic pathways:

- Uridine disphosphate glucose
- Uridine diphosphate glucuronic acid
- Cytidine diphosphate diacylglycerol

# Chemistry


Nucleotides are made up of:
A nitrogenous base
A sugar
A phosphate

The nitrogenous base combines with sugar to form a nucleoside:

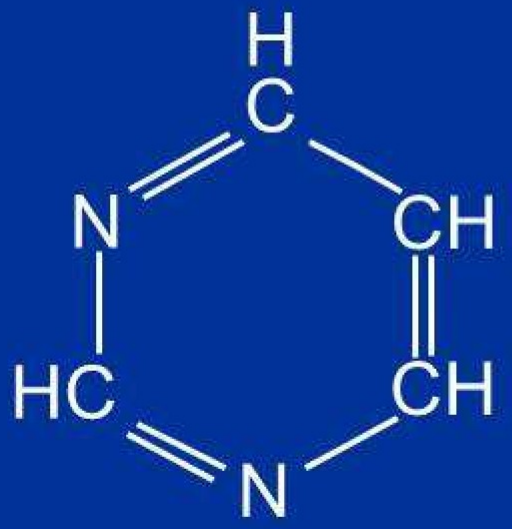
Base + Sugar → Nucleoside

The nucleoside combines with phosphate to form a nucleotide:

Nucleoside + Phosphate → Nucleotide

# Nitrogenous bases

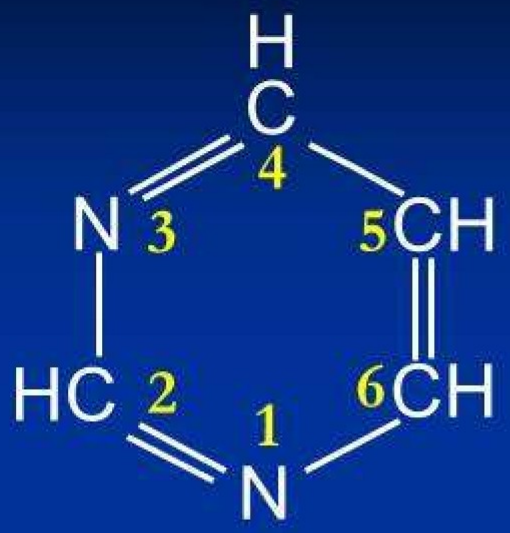
Two types of nitrogenous bases are found in nucleotides




Pyrimidine




Purine




Pyrimidine




Purine

Numbering of carbon and nitrogen atoms

The major pyrimidine bases in nucleotides are:

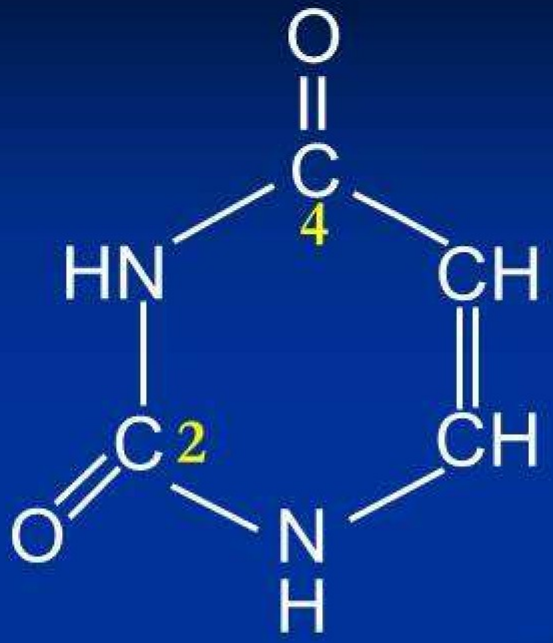
- Cytosine
- Uracil
- Thymine

The major purine bases in nucleotides are:

- Adenine
- Guanine




Cytosine
(2-Oxy-4-amino-pyrimidine)




Uracil
(2,4-Dioxy-pyrimidine)




Thymine
(2,4-Dioxy-5-methylpyrimidine)

# Pyrimidine bases




Adenine
(6-Aminopurine)



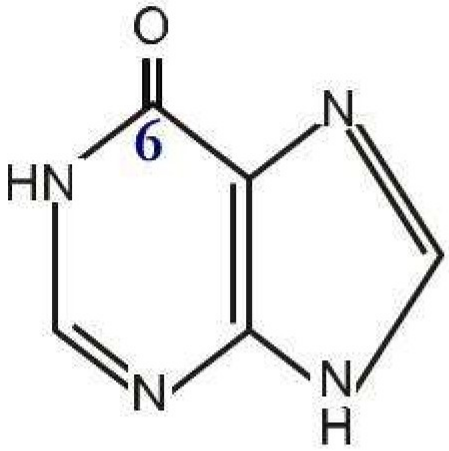

Guanine
(2-Amino-6-oxypurine)

Purine bases

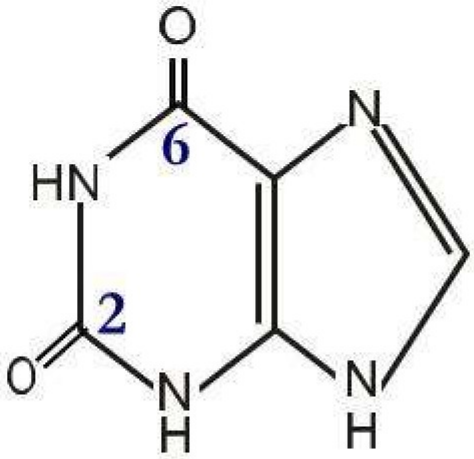
Hypoxanthine and xanthine, are metabolites of adenine and guanine

These are ultimately converted into uric acid

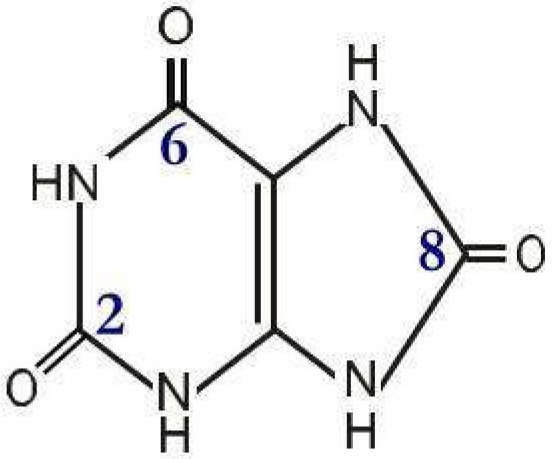
Uric acid is the end product of purine catabolism




Hypoxanthine
(6-Oxypurine)




Xanthine
(2,6-Dioxypurine)




Uric acid
(2,6,8-Trioxypurine)

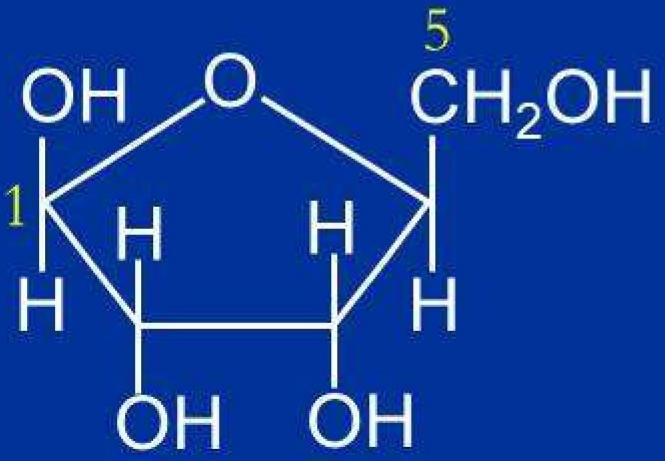
# Metabolites of purines

Some unusual bases found in nucleic acids, specially RNA:

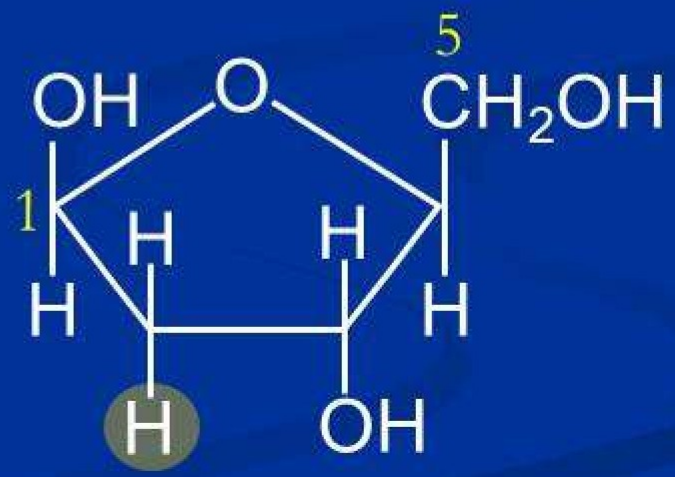
- $N^6$-Methyladenine
- $N^6$-Dimethyladenine
- $N^7$-Methylguanine
- 5-Methylcytosine
- 5-Hydroxymethylcytosine

# Sugars

Two pentose sugars are found in nucleotides




Ribose




Deoxyribose

# Nucleosides

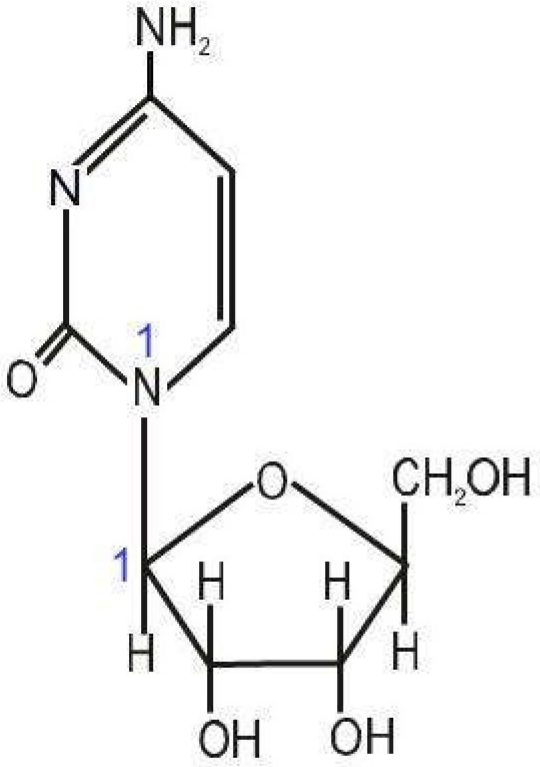
The sugar is linked to a base to form a nucleoside

A carbon-nitrogen bond is formed between:

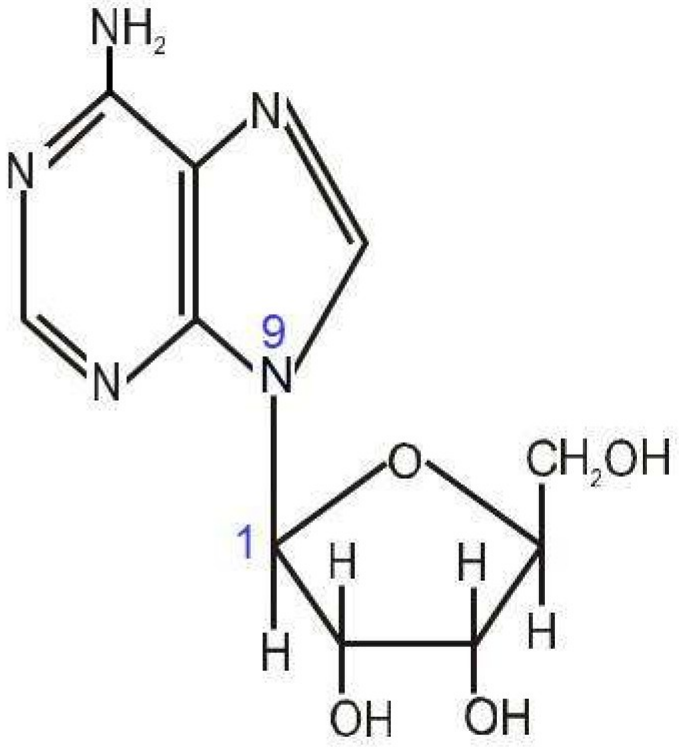
Carbon atom 1 of sugar and nitrogen atom 1 of the pyrimidine base

Carbon atom 1 of sugar and nitrogen atom 9 of the purine base

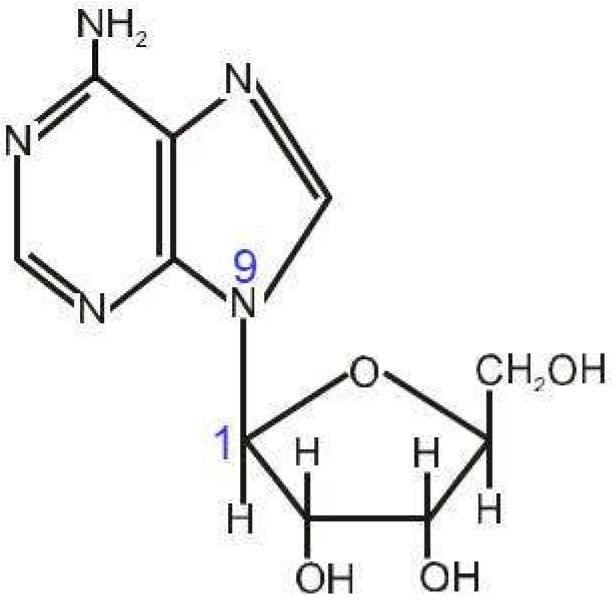
# Nucleosides



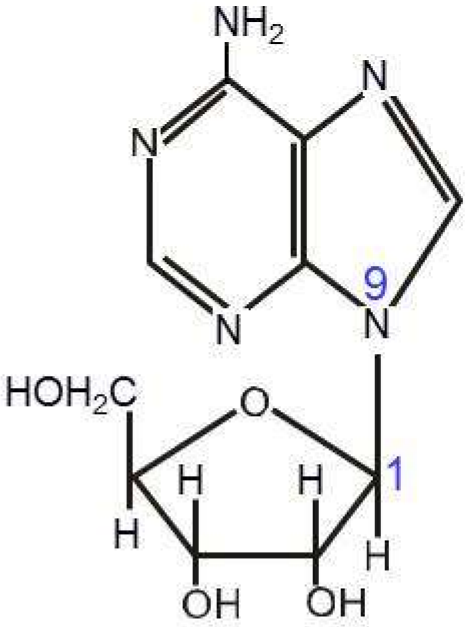

Cytidine
(cytosine + ribose)




Adenosine
(adenine + ribose)




*anti* conformation
(base and sugar are on opposite sides of the C–N bond)




*syn* conformation
(base and sugar are on same side of the C–N bond)

The *anti* conformation is more common

# Nucleosides containing ribose are:

- Cytidine (cytosine + ribose)
- Uridine (uracil + ribose)
- Adenosine (adenine + ribose)
- Guanosine (guanine +

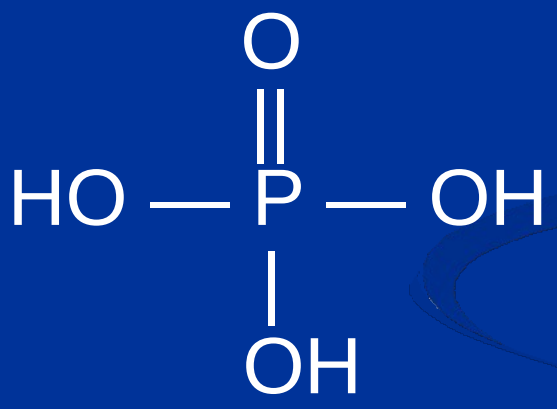
Nucleosides containing deoxyribose are:

- Deoxycytidine (cytosine+deoxyribose)
- Deoxythymidine (thymine+deoxyribose)
- Deoxyadenosine
- (adenine+deoxyribose)

Deoxyguanosine (guanine+deoxyribose)

# Phosphoric acid

The nucleosides combine with phosphoric acid ($H_3PO_4$) to form nucleotides




Phosphoric acid

# Nucleotides

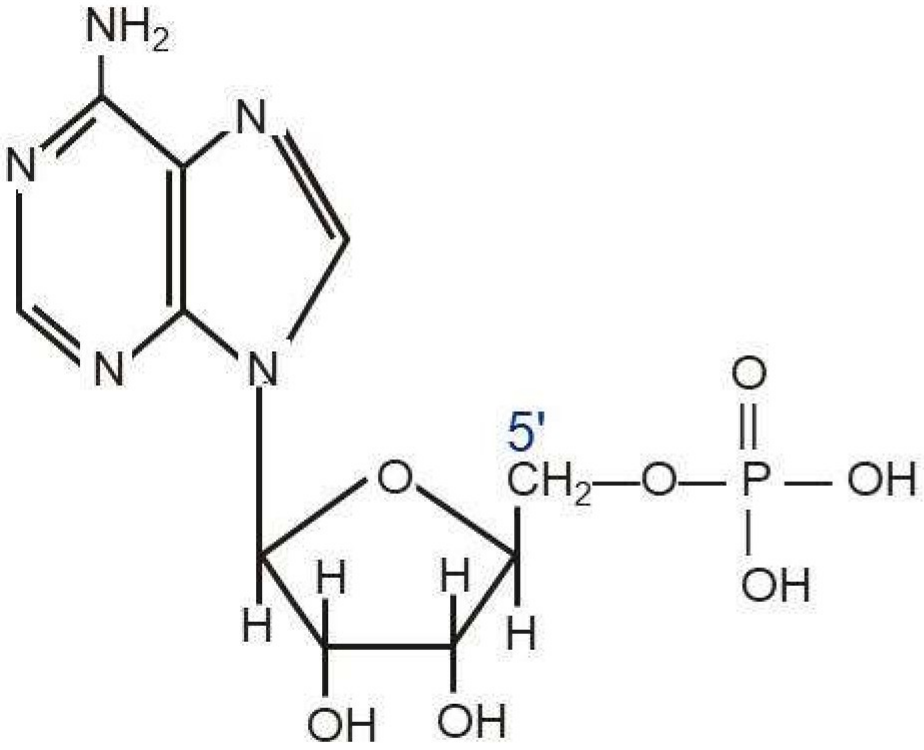
One of the –OH groups of the sugar portion of nucleoside is esterified with phosphoric acid

The phosphate group is generally bound to carbon atom 3 or 5 of the sugar

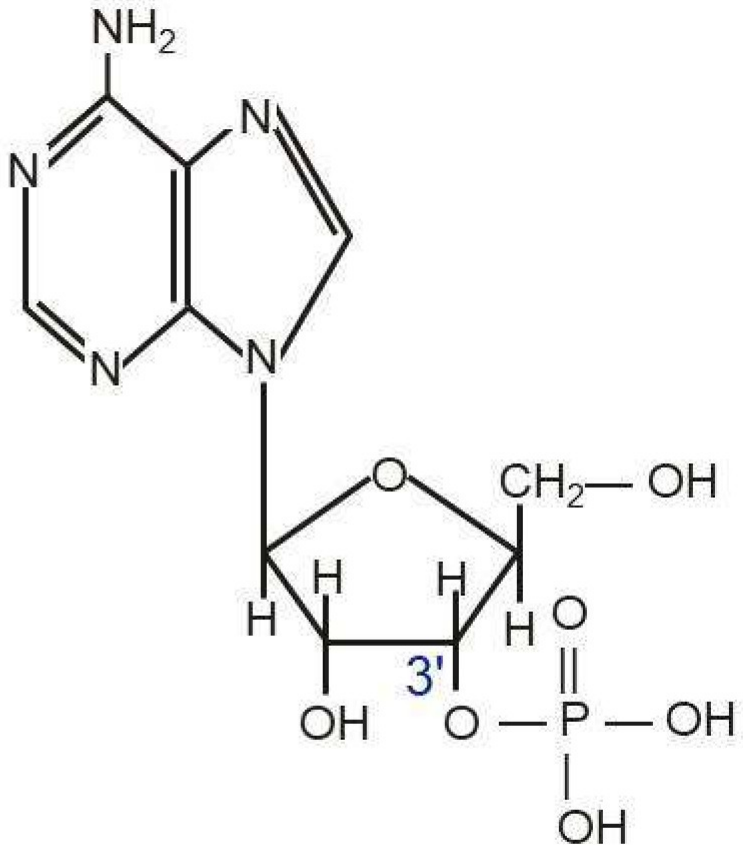
# Numbering of atoms

To avoid confusion between numbering of various atoms in the base and the carbon atoms of the sugar, those of the latter are numbered 1', 2', 3' etc

The prime mark on the numeral shows that it is carbon atom of the sugar




Adenosine-5’-monophosphate




Adenosine-3’-monophosphate

# Abbreviations

A single-letter abbreviation is commonly used for ribonucleosides

Adenosine, guanosine, cytidine and uridine are shown as A, G, C and U respectively

Deoxyadenosine, deoxyguanosine, deoxy- cytidine and deoxythymidine are shown as dA, dG, dC and dT respectively

A single phosphate group, attached to the sugar portion of the nucleotide, is represented by the letters MP

| Nucleotide | Abbreviation |
| --- | --- |
| Adenosine monophosphate | AMP |
| Guanosine monophosphate | GMP |
| Cytidine monophosphate | CMP |
| Uridine monophosphate | UMP |

The letter ‘d’ precedes the abbreviations in deoxyribonucleotides

| Nucleotide | Abbreviation |
|---|---|
| Deoxyadenosine monophosphate | dAMP |
| Deoxyguanosine monophosphate | dGMP |
| Deoxycytidine monophosphate | dCMP |
| Deoxythymidine monophosphate | dTMP |

When the phosphate group is attached to carbon atom 5 of the sugar, its position is not indicated

Thus, AMP means adenosine-5'-mono-phosphate and so on

When phosphate is attached to a carbon atom other than 5, its position is indicated by a numeral

For example, A-3'-MP means adenosine- 3'-monophosphate

More than one phosphate groups are present in some nucleotides

First phosphate group is bonded to an –OH group of the sugar by an ester linkage

Subsequent phosphates are attached to preceding ones by acid anhydride bonds

When two phosphate groups are present in a nucleotide, they are denoted by the letters DP (diphosphate)

Three phosphate groups are shown by the letters TP (triphosphate)

| Nucleotide | Abbreviation |
| --- | --- |
| Adenosine monophosphate | AMP |
| Adenosine diphosphate | ADP |
| Adenosine triphosphate | ATP |

# Some important nucleotides

Many nucleotides function as free nucleotides which include:

- Adenine derivatives
- Guanine derivatives
- Hypoxanthine derivatives
- Cytosine derivatives
- Uracil derivatives
- Vitamin derivatives
- Synthetic derivatives

# Adenine derivatives

- AMP, ADP and ATP are the nucleotides containing adenine
- ADP and ATP are described as high-energy phosphates

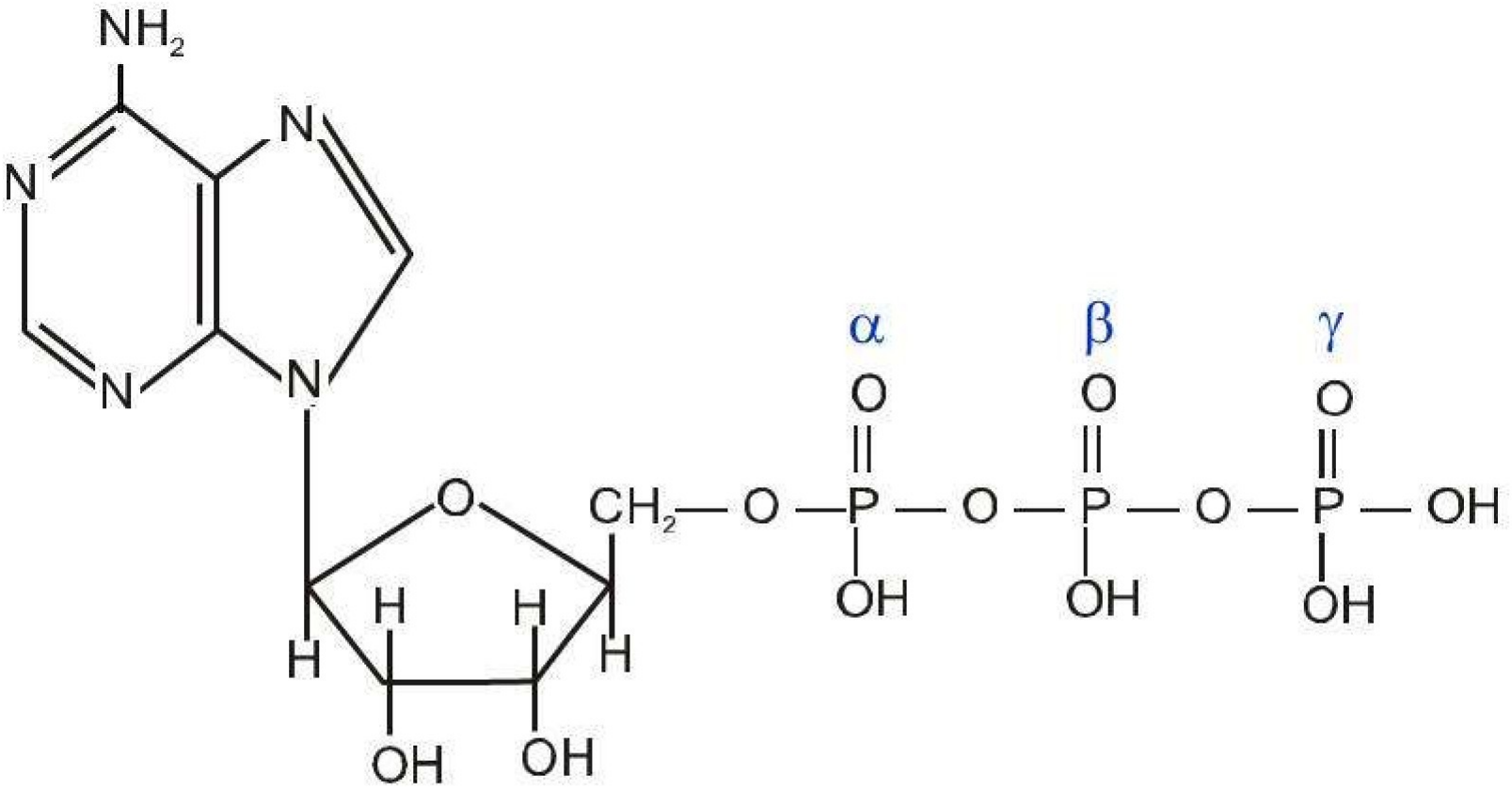
Removal of last two phosphates of ATP liberates a large amount of energy

Removal of the last phosphate of ADP liberates a large amount of energy

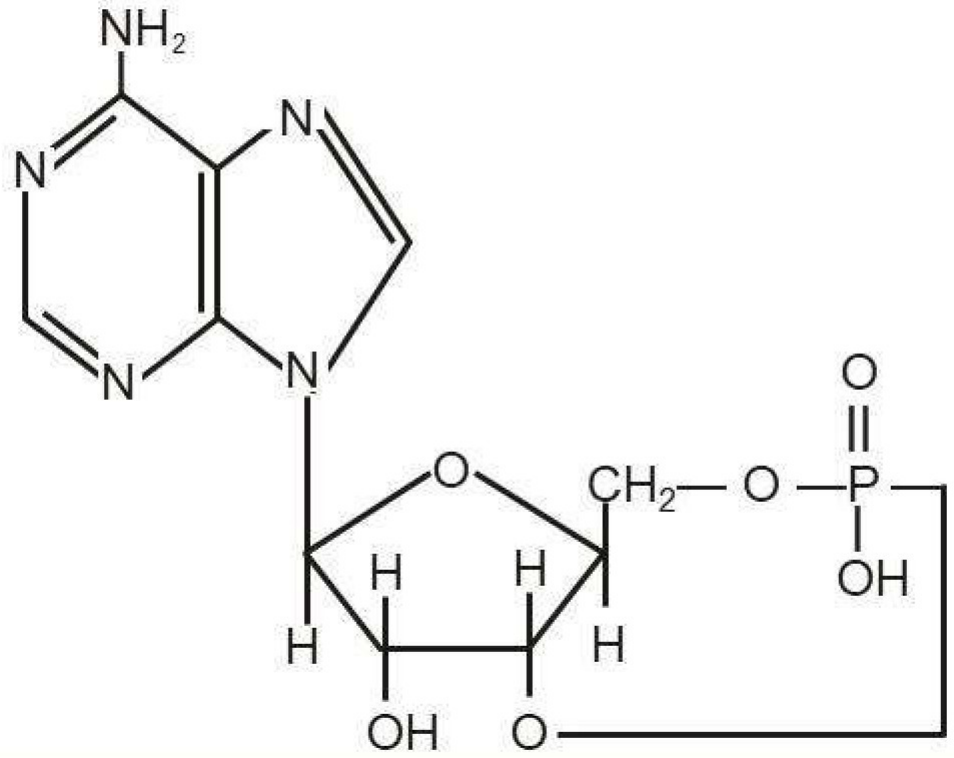
Therefore, ADP and ATP are said to be high-energy phosphates

ATP and ADP can be used as source of energy in energy-consuming reactions

During biological oxidation, energy is captured in the form of ATP by phosphorylation of ADP


NH2
N
N
N
N
O
CH2
O
P
O
OH
α
β
γ
H
H
H
H
OH
OH
Adenosine-5'-monophosphate (AMP)
Adenosine-5'-diphosphate (ADP)
Adenosine-5'-triphosphate (ATP)

Another nucleotide derived from adenine is adenosine-3', 5'-monophosphate or cyclic AMP (cAMP)




It mediates the actions of several hormones

Cyclic AMP is formed by the action of adenylate cyclase on ATP

It can be converted into AMP by another enzyme, cAMP phosphodiesterase



# Guanine derivatives

The guanine-containing nucleotides are GMP, GDP, GTP and cyclic GMP

GDP and GTP are high-energy phosphates

Cyclic GMP is a second messenger, and is also involved in the process of vision




Cyclic GMP is formed by the action of guanylate cyclase on GTP

Cyclic GMP can be converted into GMP by cGMP phosphodiesterase

# Hypoxanthine derivatives

Hypoxanthine and ribose form a nucleoside, inosine

Inosine monophosphate (IMP) is the nucleotide formed from inosine

IMP is an intermediate in

# Cytosine derivatives

- CMP, CDP and CTP are the nucleotides formed from cytosine
- CDP and CTP are high-energy phosphates
- CDP-Diacylglycerol and CDP-choline are intermediates in some pathways of lipid metabolism

# Uracil derivatives

- UMP, UDP and UTP are the uracil-containing nucleotides
- UDP and UTP are high-energy phosphates
- UDP-Glucose, UDP-galactose and UDP-glucuronic acid are intermediates in some metabolic pathways

# Vitamin derivatives

Most water-soluble vitamins perform their functions in the form of coenzymes

Many coenzymes are nucleotides containing adenine, ribose, phosphate and some other groups

Examples are flavin adenine dinucleotide (FAD), nicotinamide adenine dinucleotide (NAD) etc

# Synthetic derivatives

Some synthetic nucleosides and bases are used as drugs

They are used mainly as anti- cancer and anti-viral drugs

Cell division requires replication of cellular DNA

Replication of DNA requires a ready supply of nucleotides

Abnormal nucleotides can block replication and cell division

Substitutions can be made in the structures of purine and pyrimidine bases

Substitutions can also be made in the sugars found in nucleosides

Substituted bases or nucleosides having substituted bases/sugars can be used as drugs

If substituted bases or nucleosides are given to a subject, they are incorporated in the nucleic acids

DNA having abnormal bases or sugars cannot be replicated, and this prevents cell division

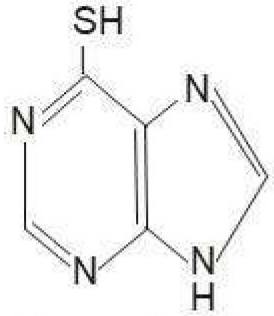
Some substituted bases/nucleosides are used as anti-cancer drugs

Some are used as antiviral drugs to check the multiplication of viruses

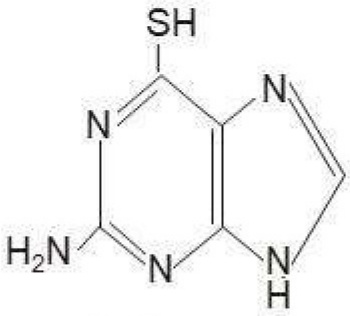
## Examples of synthetic bases and nucleosides used as anti-cancer drugs:

- 6-Mercaptopurine
- 6-Thioguanine
- 5-Fluorouracil
- 5-Iododeoxyuridine
- Cytosine arabinoside
- Adenine arabinoside

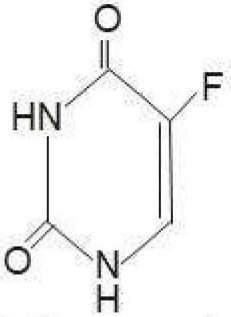
# Synthetic bases and nucleosides used as anti-cancer drugs



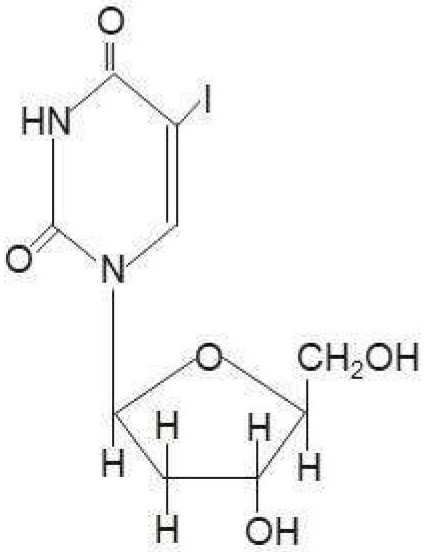

6-Mercaptopurine



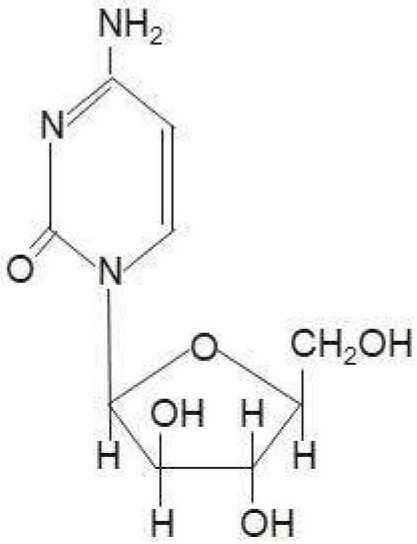

6-Thioguanine



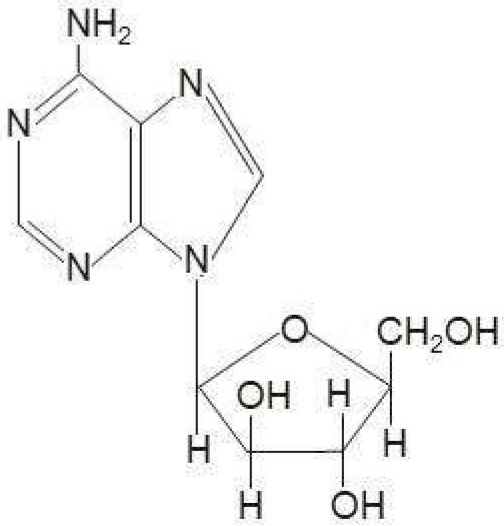

5-Fluorouracil




5-Iododeoxyuridine

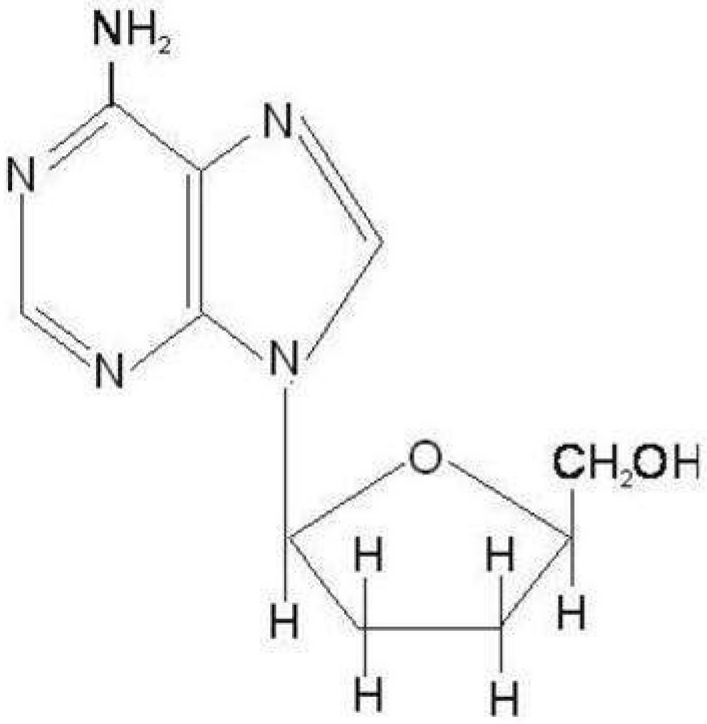



Cytosine arabinoside

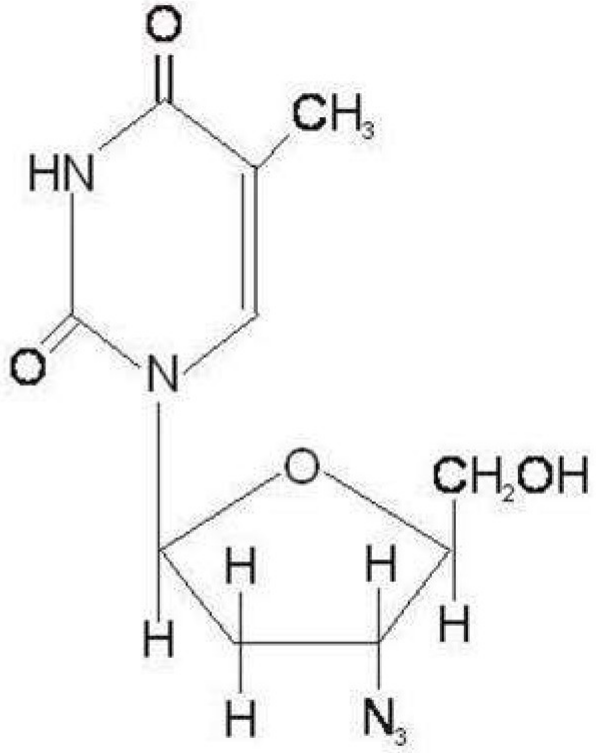



Adenine arabinoside

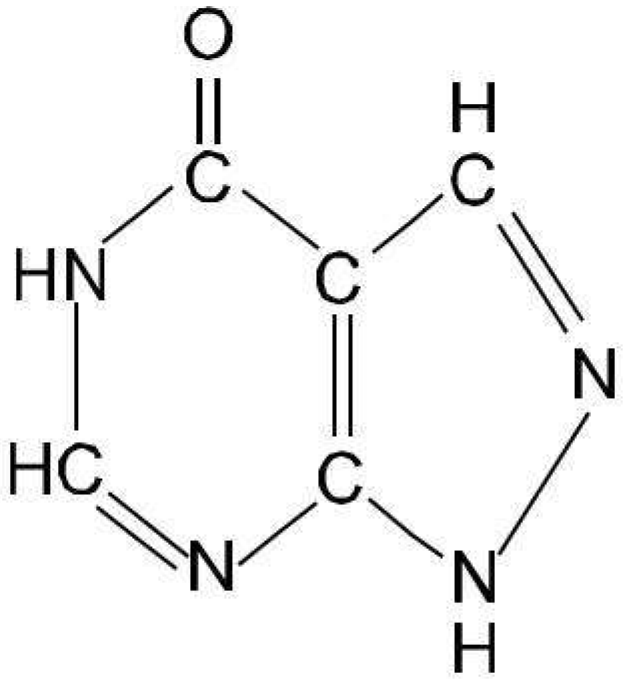
# Synthetic nucleosides used in the treatment of AIDS




Dideoxyadenosine




Azidothymidine (AZT)




Allopurinol - A structural analogue of hypoxanthine used in the treatment of gout

# Lippincott
# Chatterjea
# Harper